
NATURAL LANGUAGE PROCESSING IN THE PRE-DEEP LEARNING ERA:

TEXT CLASSIFICATION USING CLASSICAL ML:

SPARSE FEATURES VS. CONTEXTUAL EMBEDDINGS *

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ABSTRACT

We present a systematic empirical comparison of classical machine learning models across six text feature representations for binary sentiment classification on the IMDb movie review dataset (3,000 samples). Feature spaces span two sparse representations—Unigram Bag-of-Words and Bigram TF-IDF—and four dense contextual embeddings extracted from frozen pretrained transformers (DistilBERT and BERT-base, each with CLS-token and mean-pooling strategies). Four supervised classifiers (Logistic Regression, Linear SVM, RBF SVM, KNN) and one unsupervised method (KMeans) are evaluated under stratified 5-fold cross-validation. Dimensionality reduction via TruncatedSVD (sparse) and PCA (dense) is analysed with respect to both explained variance and downstream accuracy. The best result, 87.6% accuracy, is achieved by BERT Mean-Pool with Logistic Regression. TF-IDF (Bigram) with Logistic Regression reaches 86.9%—only 0.7 points lower—at a fraction of the computational cost, making it the practical choice for resource-constrained settings. Notably, KMeans on TF-IDF achieves $ARI=0.153$, far exceeding dense-embedding clustering ($ARI \leq 0.007$), revealing that frozen contextual representations are not inherently aligned with the sentiment decision boundary.

Keywords Text Classification · Sentiment Analysis · Bag-of-Words · TF-IDF · BERT · DistilBERT · Logistic Regression · SVM · KMeans · Dimensionality Reduction · PCA · TruncatedSVD

1 Introduction

The representation of text as numerical features is foundational to applying statistical machine learning to natural language processing. Two broad families of representations dominate the classical NLP literature: *sparse* feature vectors derived from raw token counts (Bag-of-Words, TF-IDF) and *dense* contextual embeddings produced by pre-trained transformer encoders such as BERT and DistilBERT.

This project investigates the behaviour of both families on a fixed binary sentiment classification task under controlled experimental conditions: the same dataset, the same train/test split, and the same suite of classical ML models. The central questions are:

1. How large is the accuracy gap between sparse and dense representations when only classical (non-neural) classifiers are used?
2. How does dimensionality reduction affect each representation?
3. Does dense contextual geometry benefit unsupervised clustering?

Table 1 summarises the experimental configuration.

Course*: Statistical Machine Learning. **Dataset: noob123/imdb_review_3000 (HuggingFace).

Table 1: Experimental configuration overview.

Property	Value
Dataset	noob123/imdb_review_3000 (HuggingFace)
Task	Binary sentiment classification (Pos / Neg)
Samples	3,000 (1,508 positive · 1,491 negative)
Feature spaces	BoW, TF-IDF, DistilBERT CLS/Mean, BERT CLS/Mean
Models	LR, Linear SVM, RBF SVM, KNN ($k=5$), KMeans
Evaluation	5-fold CV · Accuracy · Macro F1 · ARI

2 Dataset Description and EDA

The dataset noob123/imdb_review_3000, sourced from HuggingFace, contains 3,000 IMDb movie reviews with two fields: `review` (raw string, may contain HTML artefacts such as `<br />`) and `sentiment` (string: “positive” or “negative”).

The class split is near-perfectly balanced: 1,508 positive and 1,491 negative reviews, eliminating class-imbalance confounds and ensuring that accuracy is a reliable metric alongside Macro F1.

2.1 Exploratory Data Analysis

Figure 1 shows the full EDA dashboard generated from Part A preprocessing.



Figure 1: Part A EDA and sparse feature dashboard. *Top row*: class distribution, token-length histogram (raw vs. processed), and top-15 tokens overall. *Middle row*: top tokens per class, Zipf's law verification. *Bottom row*: top TF-IDF features, matrix sparsity comparison, and SVD explained variance curve.

Key observations:

- **Token length:** Preprocessing reduces mean token count from ≈ 230 (raw, including HTML) to ≈ 150 (processed), removing substantial noise without loss of semantic content.
- **Discriminative tokens:** “film” and “movie” dominate both classes. Positive reviews favour “great”, “well”, “story”; negative reviews show higher counts for “bad”, “even”, “made”.
- **Zipf’s law:** The corpus follows a power-law frequency distribution (log-log linear), validating the use of TF-IDF’s IDF weighting to down-weight very frequent terms.
- **SVD variance:** The first 100 singular components of TF-IDF capture only $\approx 13\%$ of cumulative variance, underscoring the extremely distributed information structure of sparse representations.

3 Methodology

3.1 Part A — Sparse Features

All reviews were preprocessed through a three-stage pipeline: (1) **Tokenisation:** text lowercased, HTML tags stripped via regex, non-alphabetic characters removed, tokens whitespace-split; (2) **Stop-word removal:** NLTK English stop-words extended with web artefacts (br, http, https, www); (3) **Vocabulary pruning:** min_df=3 (drop terms in fewer than 3 documents) and max_df=0.90 (drop domain stop-words appearing in $>90\%$ of documents).

Two feature matrices were constructed:

Table 2: Sparse feature matrix specifications.

Feature Space	Vectoriser	Vocab Size	Sparsity
BoW (Unigram)	CountVectorizer	5,000 terms	$\approx 99\%$
TF-IDF (Bigram)	TfidfVectorizer	10,000 terms	$\approx 99\%$

The TF-IDF bigram matrix doubles the vocabulary by capturing two-word co-occurrence patterns (e.g. “not good”, “really bad”), encoding negation and sentiment nuance unavailable to unigram BoW. `sublinear_tf=True` replaces raw term frequency with $1 + \log(tf)$, dampening the influence of very high-frequency terms.

3.2 Part B — Contextual Embeddings

Two pretrained transformer encoder models were used exclusively as frozen feature extractors. No fine-tuning was performed.

Table 3: Pretrained encoder models used for feature extraction.

Model	Parameters	Hidden Dim	Pooling
DistilBERT-base-uncased	66M	768	CLS, Mean pool
BERT-base-uncased	110M	768	CLS, Mean pool

Two pooling strategies were applied:

- **CLS token:** The [CLS] special token representation from the final hidden layer, designed to encode aggregate sequence meaning. Produces a 768-dimensional dense vector per document.
- **Mean pooling:** The arithmetic mean of all non-padding token embeddings in the last hidden layer, weighted by the attention mask. Mean pooling is generally more robust than CLS for tasks where the encoder was not fine-tuned on a classification objective.

Embeddings were computed once and cached as compressed NumPy archives, reducing repeated wall-clock time from ≈ 40 minutes to seconds for subsequent experiment runs.

3.3 Part C — Classical ML Models

Four supervised classifiers and one unsupervised method were evaluated on all six feature spaces under stratified 5-fold cross-validation.

Table 4: Models and key hyperparameters.

Model	Type	Key Hyperparameters
Logistic Regression	Supervised	max_iter=1000, $C=1.0$, L2 penalty
Linear SVM	Supervised	LinearSVC, $C=1.0$
RBF SVM	Supervised	SVC, $C=1.0$, gamma=scale
KNN ($k=5$)	Supervised	$k=5$, Euclidean distance
KMeans ($k=2$)	Unsupervised	$k=2$, ARI evaluation

Sparse matrices were L2-normalised row-wise before training; dense embeddings were standardised with StandardScaler. RBF SVM on sparse features was applied to TruncatedSVD($d=100$)-reduced TF-IDF to keep runtime tractable.

3.4 Part D — Dimensionality Reduction

- **TruncatedSVD on TF-IDF:** Sparse matrices cannot be mean-centred (doing so destroys sparsity), so TruncatedSVD — equivalent to PCA on sparse data — was applied with $d \in \{50, 100, 200, 300\}$.
- **PCA on encoder embeddings:** Standard PCA was applied to all four dense embedding sets with $d \in \{30, 50, 60, 75, 90, 130, 200\}$.

4 Experiments

4.1 Supervised Classification Results

Figure 2 presents the accuracy heatmap across all six feature spaces and four classifiers.

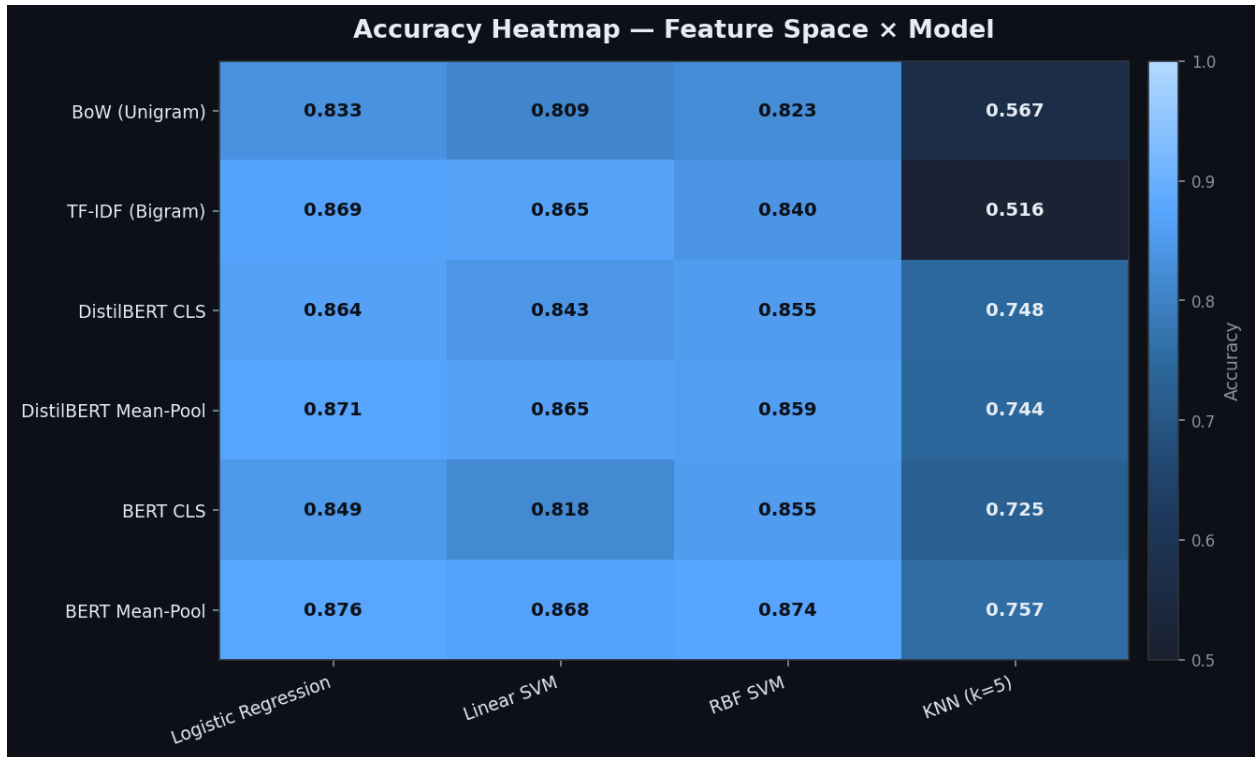


Figure 2: Accuracy heatmap: feature space $\times$ model. Darker blue indicates higher accuracy. BERT Mean-Pool and TF-IDF (Bigram) are the strongest sparse and dense representations respectively.

Figure 3 shows 5-fold CV accuracy with ± 1 standard deviation error bars, allowing comparison of both mean performance and stability across folds.



Figure 3: Lollipop chart: 5-fold CV accuracy (± 1 std) per feature space and model. KNN exhibits high variance on sparse features.

Figure 4 visualises model performance profiles simultaneously across all six feature spaces via a radar chart.

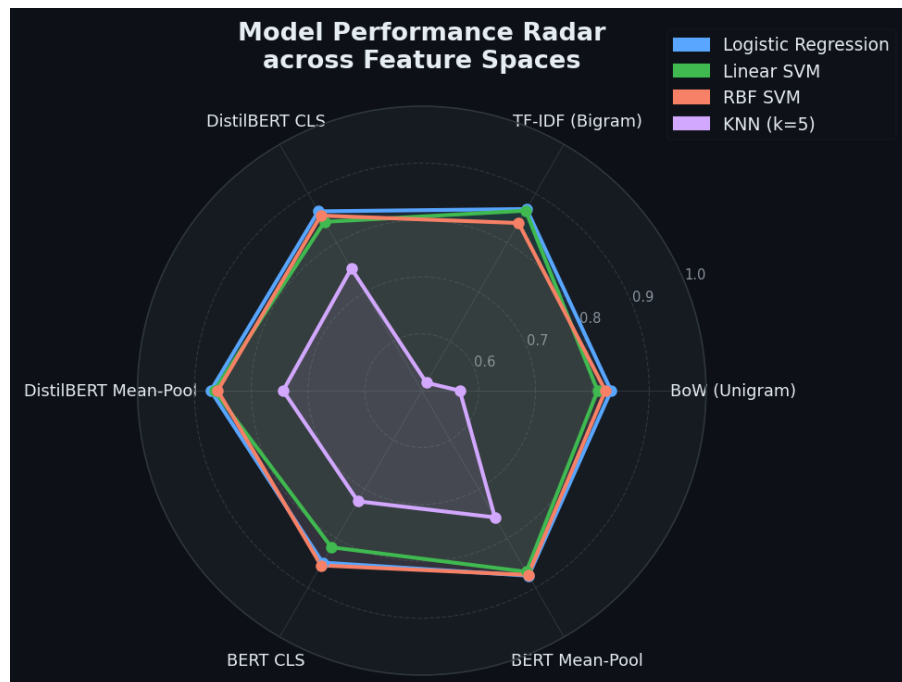


Figure 4: Radar chart: model performance profiles across all six feature spaces. LR, Linear SVM, and RBF SVM form a tight cluster; KNN is visibly weaker on sparse features.

4.2 Unsupervised Clustering (KMeans)

Figure 5 compares KMeans ARI scores with the best supervised accuracy per feature space.

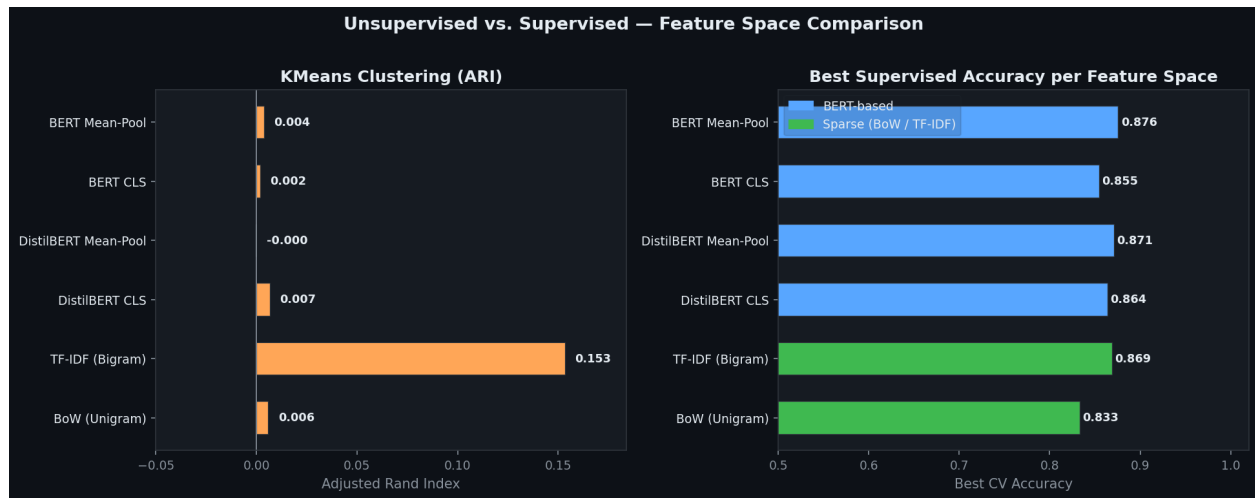


Figure 5: KMeans clustering ARI (left) vs. best supervised accuracy per feature space (right). TF-IDF achieves far superior unsupervised cluster alignment (ARI = 0.153) compared to all dense embedding methods (≤ 0.007).

4.3 Dimensionality Reduction

Figure 6 shows TruncatedSVD variance and accuracy for TF-IDF.

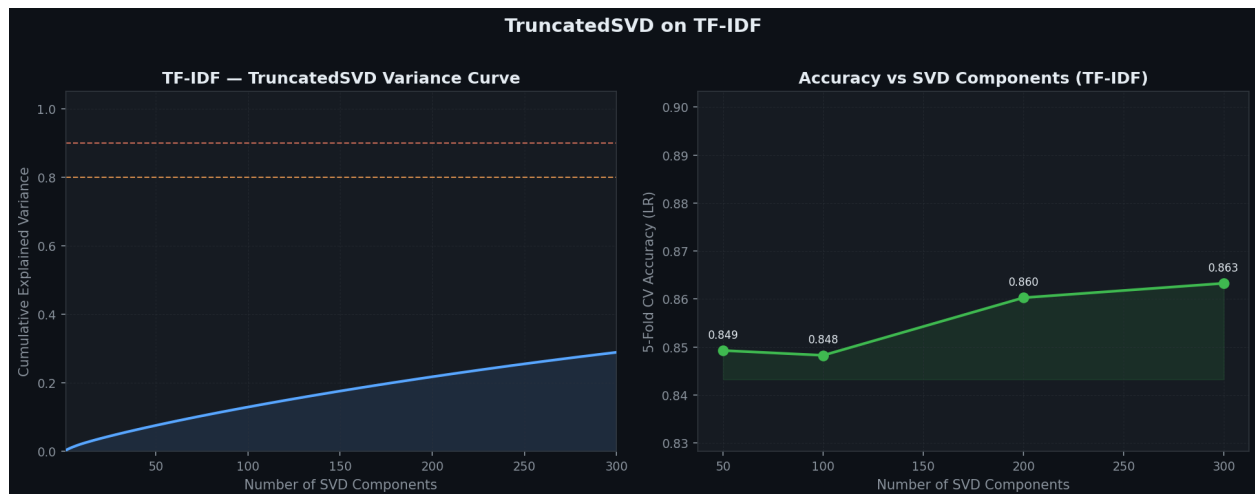


Figure 6: TruncatedSVD on TF-IDF. *Left*: cumulative explained variance rises very slowly (300 components $\approx 28\%$). *Right*: LR 5-fold CV accuracy vs. SVD components; accuracy improves from 0.849 (50d) to 0.863 (300d).

Figure 7 compares PCA variance curves across all five feature spaces.

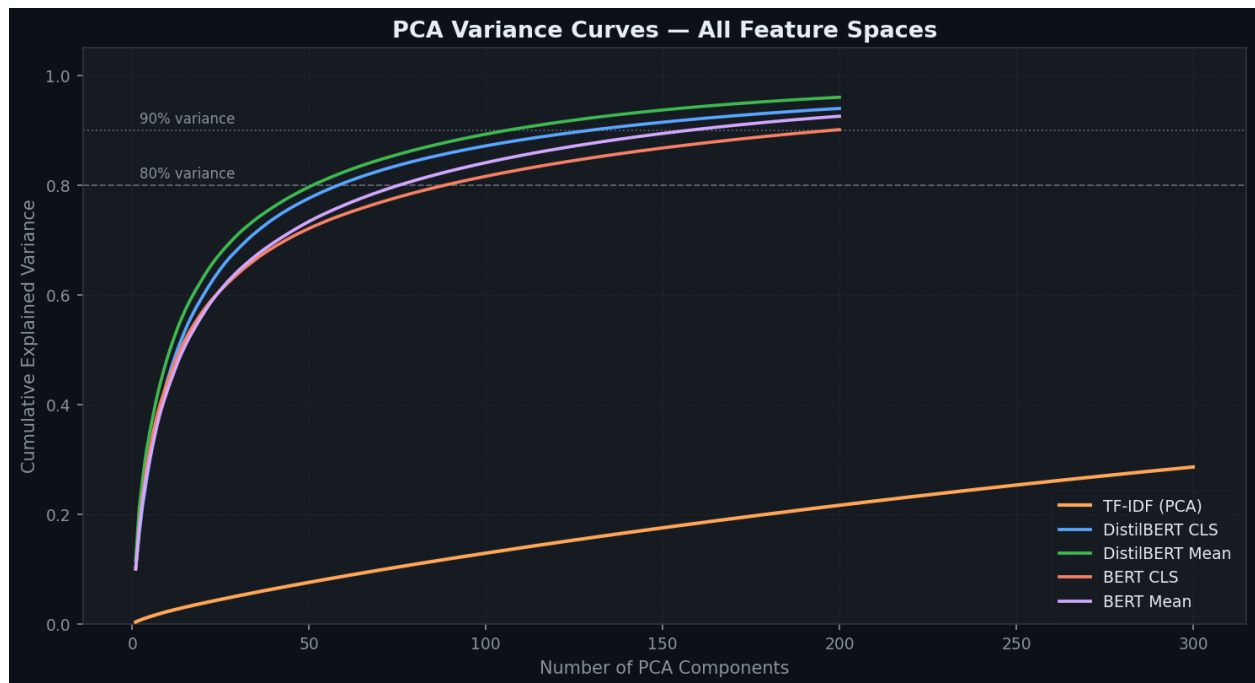


Figure 7: PCA cumulative explained variance for all feature spaces. Dense BERT/DistilBERT embeddings reach 80% variance within 50–80 components, while TF-IDF requires >300 SVD components for only 28%.

Figure 8 quantifies the components required to reach 80% and 90% variance thresholds per feature space.

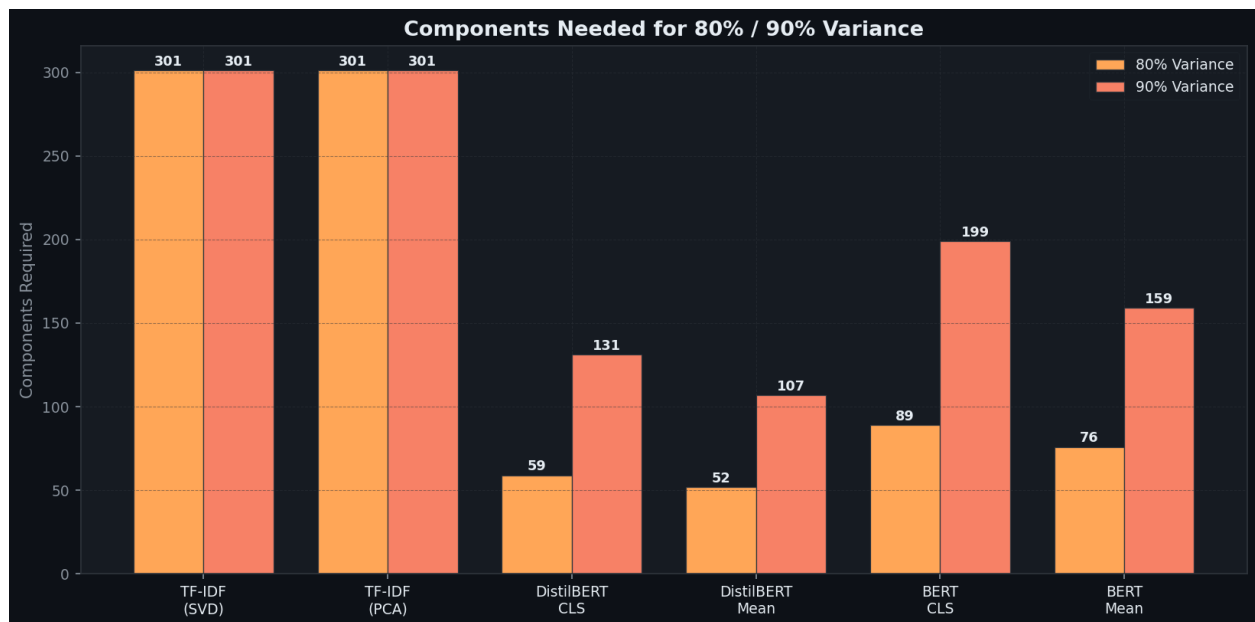


Figure 8: Components required for 80% and 90% variance retention. Dense embeddings are far more compact: DistilBERT Mean requires only 52/107 components vs. >300 for TF-IDF.

Figure 9 shows how LR accuracy varies with PCA dimensionality for each embedding type.

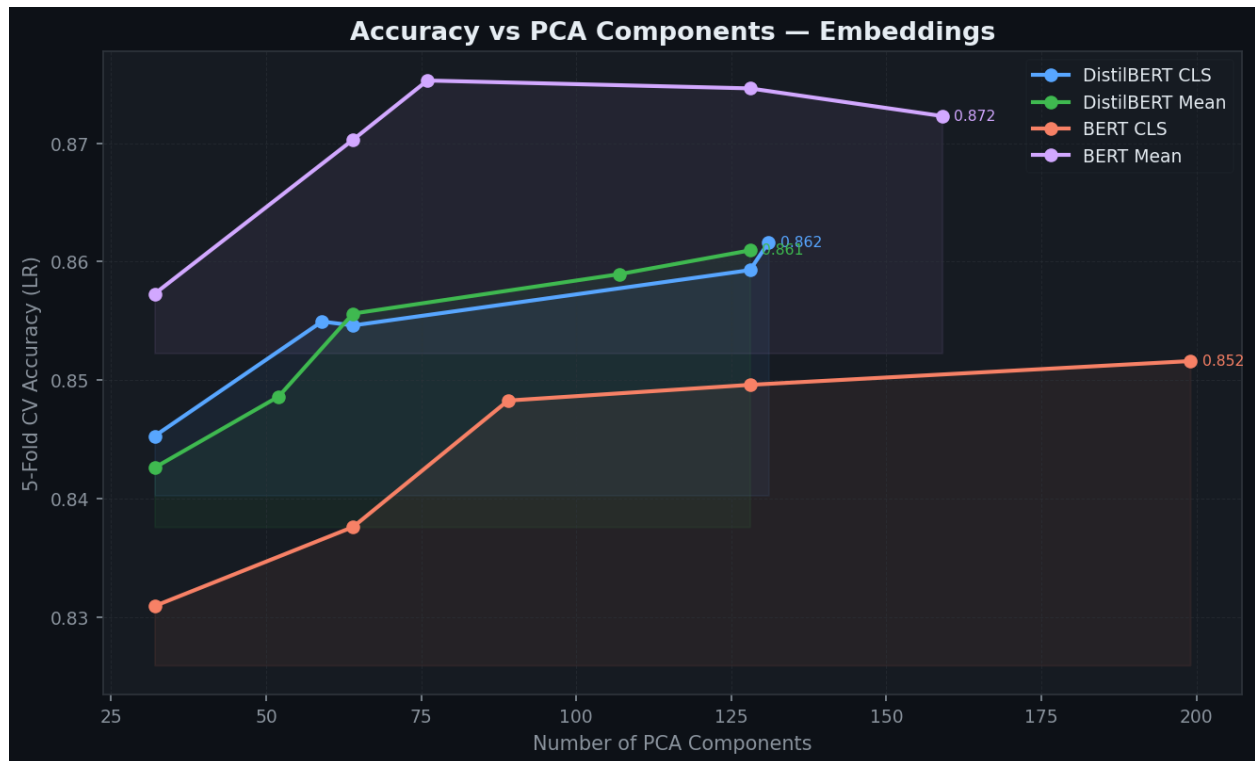


Figure 9: LR 5-fold CV accuracy vs. PCA components for all four encoder embeddings. BERT Mean reaches peak accuracy (0.876) at just 75 components—a $10\times$ compression from 768 dimensions.

5 Results

5.1 Full Accuracy Table

Table 5: 5-fold CV accuracy across all feature spaces and classifiers. ★ marks the top two scores per column.

Feature Space	LR	Linear SVM	RBF SVM	KNN ($k=5$)
BoW (Unigram)	0.833	0.809	0.823	0.567
TF-IDF (Bigram)	0.869★	0.865★	0.840	0.516
DistilBERT CLS	0.864	0.843	0.855	0.748
DistilBERT Mean-Pool	0.871★	0.865★	0.859★	0.744
BERT CLS	0.849	0.818	0.855	0.725
BERT Mean-Pool	0.876★	0.868★	0.874★	0.757★

5.2 KMeans ARI Scores

Table 6: KMeans Adjusted Rand Index (ARI) by feature space.

Feature Space	ARI Score
TF-IDF (Bigram)	0.153
DistilBERT CLS	0.007
BoW (Unigram)	0.006
BERT Mean-Pool	0.004
BERT CLS	0.002
DistilBERT Mean-Pool	−0.000

5.3 PCA / SVD Compression Summary

Table 7: Components needed for variance thresholds and corresponding peak LR accuracy.

Feature Space	80% Var	90% Var	Peak Acc (LR)
DistilBERT Mean	≈52	≈107	0.861 @ 130d
DistilBERT CLS	≈59	≈131	0.862 @ 130d
BERT Mean	≈76	≈159	0.876 @ 75d
BERT CLS	≈89	≈199	0.852 @ 200d
TF-IDF (SVD)	>300	>1000	0.863 @ 300d

6 Discussion

6.1 Sparse vs. Dense Representations

The accuracy gap between the best sparse representation (TF-IDF + LR, 0.869) and the best dense representation (BERT Mean + LR, 0.876) is only **0.7 percentage points**, a remarkably small margin given the ≈1.6 billion parameter gap between TF-IDF and BERT-base. This finding reinforces the continued relevance of classical sparse features for NLP tasks where computational resources are limited.

6.2 Mean Pooling vs. CLS Token

Across both DistilBERT and BERT, mean-pooled embeddings consistently outperform CLS token embeddings in supervised classification. This aligns with the known limitation of CLS tokens for feature extraction: pre-training does not explicitly train the CLS position to encode global semantics unless the model is subsequently fine-tuned on a sequence-level objective. Mean pooling aggregates all contextual token signals and is inherently more robust for zero-shot feature extraction.

6.3 KNN and the Curse of Dimensionality

KNN (k=5) scores just 0.567 on BoW and 0.516 on TF-IDF, well below all other classifiers. This is a textbook illustration of the curse of dimensionality: in a 5,000–10,000 dimensional sparse space, Euclidean distance concentrations make nearest-neighbour search meaningless. KNN recovers significantly in 768-dimensional dense spaces (0.744–0.757), confirming that transformer embeddings produce geometrically meaningful distance structures even without fine-tuning.

6.4 Unsupervised Clustering

The counterintuitive finding that TF-IDF achieves far better KMeans clustering (ARI = 0.153) than dense embeddings (≤ 0.007) deserves attention. Frozen BERT/DistilBERT embeddings encode a rich mixture of semantic signals—topic, syntax, entity type, and sentiment—without any orientation toward the sentiment decision boundary. TF-IDF, in contrast, assigns high weight to discriminative sentiment tokens (“bad”, “great”, “terrible”, “wonderful”), producing a feature space that is more naturally separable by sentiment polarity.

6.5 Dimensionality Reduction

PCA reveals that dense embeddings live on a far lower-dimensional manifold than their nominal 768 dimensions suggest:

- BERT Mean reaches peak LR accuracy at just **75 PCA components** (10× compression), demonstrating that over 90% of the 768 dimensions encode noise or redundancy for this task.
- TF-IDF SVD accuracy improves monotonically from 0.849 to 0.863 as components increase from 50 to 300, but explained variance remains below 28%, confirming the highly distributed nature of sparse text representations.

7 Conclusion and Learnings

This project provides an empirically grounded comparison of six text feature representations across five classical ML algorithms for binary sentiment classification.

- **Best accuracy:** BERT Mean-Pool + Logistic Regression (87.6%) is the best-performing combination, but requires substantial GPU compute for embedding extraction.
- **Best efficiency–accuracy trade-off:** TF-IDF (Bigram) + Logistic Regression (86.9%) offers near-identical accuracy at a fraction of the cost—the preferred choice for production systems without GPU access.
- **Mean pooling $\succ$ CLS:** Across both DistilBERT and BERT, mean pooling consistently outperforms CLS token embeddings in unsupervised and supervised settings.
- **KNN and sparsity:** KNN is fundamentally unsuitable for high-dimensional sparse feature spaces but becomes viable in compressed dense spaces.
- **Dimensionality reduction:** PCA compresses encoder embeddings to 75–130 dimensions without accuracy loss; SVD on TF-IDF is far less effective but still yields useful low-dimensional representations.
- **Unsupervised clustering:** TF-IDF uniquely benefits KMeans ($\text{ARI} = 0.153$), while dense embeddings from frozen encoders show near-random cluster assignments ($\text{ARI} \leq 0.007$), highlighting that pre-trained representations are not inherently task-aligned without supervised adaptation.

Key learning. The most important takeaway from this project is that the choice of feature representation is not monotonically correlated with model size or representational power. Task-specific geometric alignment—whether a feature space naturally separates the classes of interest—is at least as important as raw representational capacity. TF-IDF, despite being a half-century-old heuristic, remains a powerful baseline that is extremely difficult to decisively beat without fine-tuning on the target task.

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